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Validation of the PediBIRN-7 clinical prediction rule for pediatric abusive head trauma

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ABSTRACT

Background: The PediBIRN-7 clinical prediction rule incorporates the (positive or negative) predictive contributions of completed abuse evaluations to estimate abusive head trauma (AHT) probability after abuse evaluation. Applying definitional criteria as proxies for AHT and non-AHT ground truth, it performed with sensitivity 0.73 (95 % CI: 0.66–0.79), specificity 0.87 (95 % CI: 0.82–0.90), and ROC-AUC 0.88 (95 % CI: 0.85–0.92) in its derivation study.
Objective: To validate the PediBIRN-7's AHT prediction performance in a novel, equivalent, patient population.
Participants and settings: Consecutive, acutely head-injured children <3 years hospitalized for intensive care across eight sites between 2017 and 2020 with completed skeletal surveys and retinal exams ($N = 342$).
Methods: Secondary analysis of an existing, cross-sectional, prospective dataset, including assignment of patient-specific estimates of AHT probability, calculation of AHT prediction performance measures (ROC-AUC, sensitivity, specificity, predictive values), and completion of sensitivity analyses to estimate best- and worst-case prediction performances.
Results: Applying the same definitional criteria, the PediBIRN-7 performed with sensitivity 0.74 (95 % CI: 0.66–0.81), specificity 0.77 (95 % CI: 0.70–0.83), and ROC-AUC 0.83 (95 % CI: 0.78–0.88). The reduction in ROC-AUC was statistically insignificant ($p = .07$). Applying physicians' final consensus diagnoses as proxies for AHT and non-AHT ground truth, the PediBIRN-7 performed with sensitivity 0.73 (95 % CI: 0.66–0.79), specificity 0.87 (95 % CI: 0.82–0.90),

Abbreviations: AHT, abusive head trauma; CT, computed tomography; CI, confidence interval; LR, likelihood ratio; MRI, magnetic resonance imaging; NPV, negative predictive value; PediBIRN, pediatric brain injury research network; PPV, positive predictive value; ROC-UAC, area under the receiver operating characteristics curve.

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and ROC-AUC 0.90 (95 % CI: 0.87–0.94). Sensitivity analyses demonstrated minimal changes in rule performance.
Conclusion: The PediBIRN-7's overall AHT prediction performance has been validated in a novel, equivalent, patient population. Its patient-specific estimates of AHT probability can inform physicians' AHT-related diagnostic reasoning after abuse evaluation.

1. Introduction

To minimize missed or unrecognized cases of pediatric abusive head trauma (AHT), Pediatric Brain Injury Research Network (PediBIRN) investigators developed two screening tools (Hymel et al., 2013; Hymel et al., 2022). The PediBIRN-3 and PediBIRN-4 clinical decision rules are simple, evidence-based, and effective. Both were designed to inform physicians' early decisions to launch (or forgo) abuse evaluations in their young, acutely head-injured patients. Applied accurately and consistently, they each cast a wide net, missing ≤7 % of cases. Applied not as decision rules but as prediction tools ("AHT probability calculators"), they each facilitate patient-specific estimation of abuse probability highly predictive of the overall positive yield of patients' subsequent, completed, abuse evaluations. Their AHT screening performances have been validated externally, in multiple clinical settings, applying diverse criteria or methods to define AHT, using data captured by three independent research networks (Hymel et al., 2014; Hymel et al., 2022; Pfeiffer et al., 2018).

To estimate AHT probability after completing abuse evaluations, PediBIRN investigators derived the PediBIRN-7—a seven variable clinical prediction rule that incorporates the contributions of patients' completed skeletal surveys and retinal exams (Table 1) (Hymel et al., 2019). Patient-specific estimates of AHT probability—and the likelihood ratios (LR) that underlie each estimate—have been published for 72 unique combinations of its seven predictor variables (eTables 1 and 2; also available at www.pedibirn.com).

Applying a priori definitional criteria as proxies for AHT and non-AHT ground truth and a >0.50 probability threshold to classify patients as AHT, the PediBIRN-7 performed with sensitivity 0.73 (95 % CI: 0.66–0.79), specificity 0.87 (95 % CI: 0.82–0.90), positive predictive value (PPV) 0.79 (95 % CI: 0.72–0.85), and negative predictive value (NPV) 0.82 (95 % CI: 0.77–0.86) in its derivation study. The area under the receiver operating characteristics curve (ROC-AUC) was 0.88 (95 % CI: 0.85–0.92). Applying physicians' final consensus diagnoses as proxies for AHT and non-AHT ground truth and the same probability threshold, it performed with sensitivity 0.87 (95 % CI: 0.83–0.91), specificity 0.90 (95 % CI: 0.85–0.93), PPV 0.90 (95 % CI: 0.86–0.94) and NPV 0.87 (95 % CI: 0.82–0.91). The ROC-AUC was 0.94 (95 % CI: 0.91–0.96) (Hymel et al., 2019).

Unlike the PediBIRN 3- and 4-variable screening tools, the AHT prediction performance of the PediBIRN-7 has not been validated. Our objective was to validate the AHT prediction performance of the PediBIRN-7 in a new, equivalent, patient population, applying the same two proxies for AHT and non-AHT ground truth.

2. Methods

2.1. Overview

This was a retrospective, secondary analysis of de-identified, prospective, cross-sectional data captured by PediBIRN investigators conducting a cluster randomized clinical trial of the PediBIRN-4, with detailed methods described previously (Hymel et al., 2021). All participating sites obtained approval for the parent study with waivers of informed parental consent from their local institutional review boards. The Institutional Review Board at Penn State Health Hershey Medical Center determined that this secondary analysis of an existing, de-identified data set did not require additional review or its formal approval.

2.2. The PediBIRN dataset

The data used to derive the PediBIRN-7 was captured between 2010 and 2013 across 18 participating sites. The data used in this secondary analysis was captured between 2017 and 2020 across eight of the 18 participating sites, utilizing patient-specific data forms

Table 1
The PediBIRN-7: Seven variables used to estimate AHT probability after abuse evaluation in acutely head-injured children <3 years hospitalized for intensive care.

- Any clinically significant respiratory compromise at the scene of injury, during transport, in the Emergency Department, or prior to admission
- Any bruising involving the child's ear(s), neck, or torso
- Any subdural hemorrhage(s) or fluid collection(s) that are bilateral OR involve the interhemispheric space
- Any skull fracture(s) other than an isolated, unilateral, nondiastatic, linear, parietal skull fracture
- Skeletal survey that revealed rib fracture(s), classic metaphyseal lesion fracture(s), epiphyseal separation(s), fracture(s) of the scapula or sternum, fracture(s) of digit(s), vertebral body fracture(s) or dislocations, OR fracture(s) of spinous process(es)
- Retinal exam by an ophthalmologist that revealed retinoschisis OR retinal hemorrhages described as dense, extensive, covering a large surface area, and/or extending to the ora serrata
- Any brain hypoxia, ischemia, OR swelling

Abbreviations: AHT = abusive head trauma.

equivalent to those used to derive the PediBIRN 3-, 4-, and 7-variable clinical prediction rules (Hymel et al., 2021). It contains uniform, prospective, clinical, radiological, and historical data regarding 473 consecutive, acutely head-injured patients under three years of age hospitalized for intensive care of acute cranial or intracranial injuries confirmed on CT or MRI. Patients with preexisting brain abnormalities and patients head injured in a collision involving motor vehicle(s) were excluded. Child abuse evaluations were ordered and completed at the discretion of treating physicians. The study population for this secondary analysis was limited to the 342 (72 %) of 473 patients who underwent both radiological skeletal survey and retinal examination by an ophthalmologist.

2.3. Definitional criteria

In the absence of gold standards, we applied the same a priori definitional criteria for AHT, non-AHT, an abnormal skeletal survey, and an abnormal retinal exam used to derive the PediBIRN-7 (Table 2) (Hymel et al., 2019). To avoid circular reasoning, abnormal skeletal survey and retinal findings were excluded from the definitional criterion listing AHT characteristic physical findings.

2.4. Assignment of patient-specific estimates of AHT probability

Every patient in the validation study population was assigned a patient-specific estimate of AHT probability based on their specific combination of the PediBIRN-7's seven predictor variables (Table 1). The assigned, patient-specific estimates of probability (eTables 1 and 2; also available at www.pedibirn.com) were calculated during the prediction rule's derivation, with detailed methods described previously. (Hymel et al., 2019).

2.5. Estimation of prediction rule performance

Estimates of performance (sensitivity, specificity, predictive values) were calculated applying a probability threshold of >0.50 to classify patients as AHT. To characterize the PediBIRN-7's overall prediction performance, the ROC-AUC was calculated using sensitivity and specificity measurements calculated across a full spectrum of discrimination thresholds. Indeterminate cases that failed to meet a priori definitional criteria for AHT or for non-AHT (Table 2) were excluded from these primary analyses. To estimate best- and worst-case AHT prediction performances, we completed secondary, sensitivity analyses reclassifying all indeterminate cases as AHT and non-AHT, respectively.

2.6. Applying an alternate proxy for AHT and non-AHT ground truth

To provide a basis for comparison, we conducted a complete, duplicate analysis of the PediBIRN-7's prediction performance, applying equivalent methods and physicians' final consensus diagnoses of AHT and non-AHT as proxies for ground truth.

2.7. Comparing prediction rule performance in derivation vs. validation

Sensitivity, specificity, and predictive values in derivation vs. validation were compared using two sample proportion tests. ROC-AUC results were compared applying the bootstrap methods test, pROC package, R version 4.3.2.

Table 2
Definitional criteria used to derive and validate the PediBIRN-7 clinical prediction rule.

A patient's head trauma was labeled abusive IF…
• The primary caregiver[a] admitted abusive acts
• Abusive acts by the primary caregiver[a] were witnessed by an unbiased, independent observer
• The primary caregiver[a] specifically denied any head trauma, even though the pre-ambulatory child in his or her care became acutely, clearly and persistently ill with clinical signs subsequently linked to traumatic cranial injuries visible on CT or MRI
• The primary caregiver's[a] account of the child's head injury event was clearly historically inconsistent with repetition over time
• The primary caregiver's[a] account of the child's head injury event was clearly developmentally inconsistent with the child's known (or expected) gross motor skills
• Abuse evaluations revealed patterned bruising or dry contact burns, hot water immersion burns, or CT-confirmed intra-abdominal injury[b]
A patient's head trauma was labeled non-abusive IF…
• The child's head injury event was witnessed by an unbiased, independent observer who described the event as accidental (non-abusive), OR…
• The primary caregiver[a] provided an account of the child's head injury event that was both historically consistent with repetition over time and developmentally consistent with the child's known (or expected) gross motor skills…AND…abuse evaluation failed to reveal patterned bruising or dry contact burns, hot water immersion burns, or CT-confirmed intra-abdominal injury[b]
All remaining patients' head traumas were labeled indeterminate

Abbreviations: CT = computed tomography, MRI = magnetic resonance imaging, PediBIRN = pediatric brain injury research network.

[a] Defined a priori as the person responsible for the child when he or she was acutely head injured or first became clearly and persistently ill with clinical signs subsequently linked to traumatic cranial injuries visible on CT or MRI.

[b] To avoid circular reasoning, abnormal skeletal survey and retinal findings (Table 1) were excluded from this list of AHT characteristic physical findings.

3. Results

3.1. Patient classification

The distribution of patients ($N = 342$) classified as AHT, non-AHT, and indeterminate is presented online in eTable 3. Definitional criteria classified fewer patients as victims of AHT than physicians.

3.2. Estimation of prediction rule performance, applying definitional criteria

Applying definitional criteria (Table 2) and a probability threshold of >0.50 to classify patients as AHT, the PediBIRN-7 performed with sensitivity 0.74 (95 % CI: 0.66–0.81), specificity 0.77 (95 % CI: 0.70–0.83), PPV 0.73 (95 % CI: 0.65–0.80), and NPV 0.78 (95 % CI: 0.70–0.84). The ROC-AUC was 0.83 (95 % CI: 0.78–0.88) (Fig. 1). Secondary, sensitivity analyses revealed minimal changes in prediction performance if/when we assumed that all 41 indeterminate cases were victims of AHT or non-AHT, respectively (eTable 4, online).

3.3. Estimation of prediction rule performance, applying physicians' final consensus diagnoses

Applying physicians' final consensus diagnoses and the same probability threshold of >0.50 to classify patients as AHT, the PediBIRN-7 performed with sensitivity 0.79 (95 % CI: 0.72–0.84), specificity 0.84 (95 % CI: 0.77–0.90), PPV 0.87 (95 % CI: 0.81–0.91), and NPV 0.75 (95 % CI: 0.67–0.81). The ROC-AUC was 0.90 (95 % CI: 0.87–0.94) (Fig. 1). Secondary, sensitivity analyses again revealed minimal changes in prediction performance if/when we assumed that all 19 indeterminate cases were victims of AHT or non-AHT, respectively (eTable 5, online).

3.4. Comparison of prediction rule performance in derivation vs. validation, applying definitional criteria

Contingency tables comparing prediction rule performance in derivation vs. validation, applying definitional criteria as proxies for AHT and non-AHT ground truth, are presented in Table 3. Specificity decreased from derivation to validation [from 0.87 (95 % CI: 0.82–0.90) to 0.77 (95 % CI: 0.70–0.83), $p = .02$]. The decrease in ROC-AUC from derivation to validation was statistically insignificant [from 0.88 (95 % CI: 0.85–0.92) to 0.83 (95 % CI: 0.78–0.88), $p = .07$].

3.5. Comparison of prediction rule performance in derivation vs. validation, applying physicians' final consensus diagnoses

Contingency tables comparing prediction rule performance in derivation vs. validation, applying physicians' final consensus diagnoses as proxies for AHT and non-AHT ground truth, are presented in Table 4. Sensitivity and NPV decreased from derivation to validation [from 0.87 (95 % CI: 0.83–0.91) to 0.79 (95 % CI: 0.72–0.84), $p = .02$; and from 0.87 (95 % CI: 0.82–0.91) to 0.75 (95 % CI: 0.67–0.81), $p = .004$, respectively]. The decrease in ROC-AUC from derivation to validation was statistically insignificant [from 0.94 (95 % CI: 0.91–0.96) to 0.90 (95 % CI: 0.87–0.94), $p = .08$].

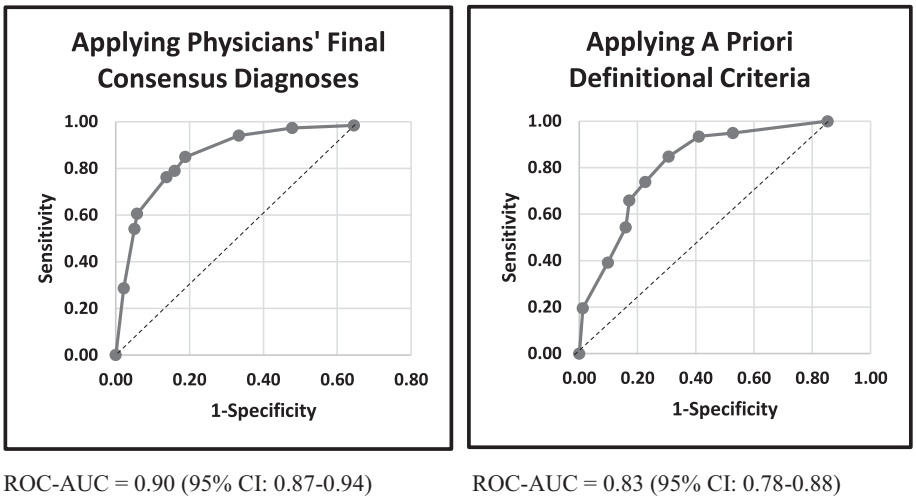

Fig. 1. Receiver operating characteristics curves, applying physicians' final consensus diagnoses and a priori definitional criteria as proxies for AHT and non-AHT ground truth, and discrimination thresholds of >0.10, >0.20, >0.30, …and >0.90 to classify patients as AHT.
ROC-AUC = 0.90 (95 % CI: 0.87–0.94) ROC-AUC = 0.83 (95 % CI: 0.78–0.88).

Table 3
Primary analyses of the PediBIRN-7's predictive performance in derivation vs. validation, applying definitional criteria to classify patients as AHT vs. non-AHT, and excluding all indeterminate cases.

PediBIRN-7 Derivation Study (N=500)			PediBIRN-7 Validation Study (N=342)			
	Applying Definitional Criteria (excluding all 44 indeterminate cases)			**Applying Definitional Criteria (excluding all 41 indeterminate cases)**		
PediBIRN-7 estimate of AHT probability	*AHT*	*Non-AHT*	PediBIRN-7 estimate of AHT probability	*AHT*	*Non-AHT*	
>0.50	136	36	*>0.50*	102	37	
≤0.50	51	233	*≤0.50*	36	126	*p values*
		Value (95% CI)			Value (95% CI)	
Sensitivity		0.73 (0.66-0.79)	Sensitivity		0.74 (0.66-0.81)	.91
Specificity		0.87 (0.82-0.90)	Specificity		0.77 (0.70-0.83)	.02
Positive predictive value		0.79 (0.72-0.85)	Positive predictive value		0.73 (0.65-0.80)	.30
Negative predictive value		0.82 (0.77-0.86)	Negative predictive value		0.78 (0.70-0.84)	.33
ROC-AUC was 0.88 (95% CI: 0.85-0.92).			ROC-AUC was 0.83 (95% CI: 0.78-0.88).			.07

Abbreviations: AHT = abusive head trauma, PediBIRN = pediatric brain injury research network, ROC-AUC = area under the receiver operating characteristics curve.

Table 4
Primary analyses of the PediBIRN-7's predictive performance in derivation vs. validation, applying physicians' final diagnoses to classify patients as AHT vs. non-AHT, and excluding all indeterminate cases.

PediBIRN-7 Derivation Study (N=500)			PediBIRN-7 Validation Study (N=342)			
	Applying Physicians' Final Diagnoses (excluding all 26 indeterminate cases)			**Applying Physicians' Final Diagnoses (excluding all 19 indeterminate cases)**		
PediBIRN-7 estimate of AHT probability	*AHT*	*Non-AHT*	PediBIRN-7 estimate of AHT probability	*AHT*	*Non-AHT*	
>0.50	216	23	*>0.50*	146	22	
≤0.50	31	204	*≤0.50*	39	116	*p values*
		Value (95% CI)			Value (95% CI)	
Sensitivity		0.87 (0.83-0.91)	Sensitivity		0.79 (0.72-0.84)	.02
Specificity		0.90 (0.85-0.93)	Specificity		0.84 (0.77-0.90)	.14
Positive predictive value		0.90 (0.86-0.94)	Positive predictive value		0.87 (0.81-0.91)	.35
Negative predictive value		0.87 (0.82-0.91)	Negative predictive value		0.75 (0.67-0.81)	.004
ROC-AUC was 0.94 (95% CI: 0.91-0.96).			ROC-AUC was 0.90 (95% CI: 0.87-0.94).			.08

Abbreviations: AHT = abusive head trauma, PediBIRN = pediatric brain injury research network, ROC-AUC = area under the receiver operating characteristics curve.

4. Discussion

The absence of a diagnostic gold standard creates a significant challenge for AHT researchers. PediBIRN investigators have routinely dealt with this challenge by completing duplicate, equivalent analyses that apply two different proxies for AHT and non-AHT ground truth: a priori definitional criteria (Table 2) and physicians' final consensus diagnoses. Each proxy has its advantages and disadvantages (Hymel et al., 2019). Definitional criteria were chosen to exclude circularity introduced by considering predictor variables in assigning outcomes, but rests substantially on physician judgements of histories that may be false while failing to consider many other variables available to diagnosing physicians. Physicians' final consensus diagnoses are likely informed by variables included in the PediBIRN-7 prediction rule and perhaps by the rule's derivation study, but consider differential diagnostic testing and non-medical data, such as assessments of psychosocial risk, witness statements, and the results of scene investigation. Our presumption has been that obtaining comparable results in duplicate, equivalent analyses conveys a measure of validity.

Replicating this approach, we completed duplicate, equivalent analyses of prediction rule performance and obtained comparable results. In both analyses, the PediBIRN-7's overall prediction performance—best represented by the ROC-AUC—was replicated in a novel patient population using equivalent research methods. Although ROC-AUC values in validation were lower than those observed in derivation, the reductions were not statistically significant.

The four completed analyses of the PediBIRN-7's prediction performance (applying two different proxies for ground truth in both derivation and validation) yielded ROC-AUC values from 0.83 (95 % CI: 0.78–0.88) to 0.94 (95 % CI: 0.91–0.96). Although these ROC-AUC results support favorable impressions of the PediBIRN-7's overall prediction performance, other specific measures of performance (sensitivity, specificity, NPV) were significantly lower in validation than in derivation (Tables 3 and 4).

Multiple explanations are possible. First, the 7-variable prediction model may have been overfitted to the data during its derivation, rendering it less able to generalize well to new data in the validation study population. Second, because the data used to derive and validate the PediBIRN-7 were captured in different time frames (2010–2013 vs. 2017–2020) and settings (18 vs. 8 participating sites), investigators' interpretation of its predictor variables, application of definitional criteria, and AHT-related diagnostic reasoning may have evolved in a way that impacted prediction rule performance over time. Finally, hospital and pediatric intensive care unit (PICU) admission practices for pediatric patients with mild head trauma likely changed over this time period, resulting in differences in case mix in the derivation vs. validation patient populations.

Because patient populations and measurements change over time, clinicians should expect heterogeneity in model performance. It follows that this single, favorable, validation study may not be sufficient to claim that the PediBIRN-7 prediction rule will work uniformly well in its intended clinical context and setting (Van Calster, Steyerberg, & van Smeden, 2023). Based on these considerations, we speculate that the PediBIRN-7's current AHT prediction performance is best represented by the performance measures demonstrated in this validation study. Moving forward, external validation studies are needed to confirm the generalizability of the model's performance over time and across diverse patient populations and settings.

In isolation, the PediBIRN-7's estimates of AHT probability should not be considered a sufficient foundation upon which to confirm or exclude AHT. Absent a gold standard, physicians caring for acutely head-injured infants and young children must continue to vigorously exclude alternate medical diagnoses before diagnosing AHT. Doing so requires active consideration of the presenting history; past and family medical history; family psychosocial risk factors; physical findings; and the results of radiologic evaluations, subspecialty consultations, scene investigations, forensic interviews, and tests that confirm or exclude medical mimics of AHT.

Based on the results of this study, we recommend that physicians also consider—both routinely and actively—their patient's specific, evidence-based estimate of AHT probability before confirming or excluding AHT. As noted in the report describing its derivation, "…physicians who find reasonable concordance between their own diagnostic impressions and the patient-specific estimates of AHT probability facilitated by the PediBIRN-7…will likely feel more confident that their impressions are valid. Physicians who instead find discordance may elect to press for further investigation, and to explore the possibilities of a false positive or negative result, an error in their clinical judgement, AHT masquerading as accidental trauma, and/or their own inexperience or implicit bias." (Hymel et al., 2019).

Routine, active consideration of the LRs that underly the PediBIRN-7's patient-specific estimates of AHT probability has additional merit. Unlike AHT probability estimates, the calculation of LRs does not depend on AHT prevalence (pre-test odds) in the study population. In addition, LRs are measures of evidential value. In this context, they each represent a ratio of two probabilities: (1) the probability of a specific set of clinical findings given that the head-injured child was abused, and (2) the probability of the same clinical findings given that the child was not abused. The LR can thus be interpreted as a numerical degree of support for the hypothesis of AHT versus the hypothesis of non-AHT. To clarify, a patient-specific LR of 20 indicates that the patient's specific combination of the prediction rule's seven (positive or negative) clinical findings (Table 1) is 20 times more likely to be observed in patients classified as AHT than in patients classified as non-AHT. Like estimates of AHT probability, physicians' routine, active consideration of the evidential value of a specific cluster of medical findings can inform their AHT-related diagnostic reasoning.

This study had strengths. The study population was large and similar in size to the derivation study population. The prospective data used to validate prediction rule performance precisely replicated that captured and used to derive the PediBIRN-7. Duplicate analyses of prediction rule performance applying two different proxies for AHT and non-AHT ground truth yielded comparable results. ROC-AUC values consistently $\geq$0.83 in both derivation and validation support favorable impressions of the PediBIRN-7's overall prediction performance.

This study also had limitations. Multiple, specific measures of prediction rule performance declined significantly from derivation to validation. The prediction model may have been overfitted to its derivation data, limiting its generalizability. The PediBIRN-7 was both derived and validated in PICU settings. Unless and until external validation studies confirm its performance in alternate clinical

settings, it should only be applied in PICU settings.

5. Conclusions

The PediBIRN-7's overall AHT prediction performance has been validated in a novel, equivalent, patient population. Its patient-specific estimates of AHT probability—and their underlying LRs—can inform physicians' AHT-related diagnostic reasoning after abuse evaluation.

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CRediT authorship contribution statement

Kent P. Hymel: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Data curation, Conceptualization. **Christopher L. Carroll:** Writing – review & editing, Investigation. **Terra N. Frazier:** Writing – review & editing, Investigation. **Kerri Weeks:** Writing – review & editing, Investigation. **Bruce E. Herman:** Writing – review & editing, Investigation. **Mark Marinello:** Writing – review & editing, Investigation. **Yiming Chen:** Writing – review & editing, Formal analysis. **Ming Wang:** Writing – review & editing, Formal analysis. **Stephen C. Boos:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors have no personal or financial conflicts of interest relevant to this article to disclose.

Data availability

All pertinent data is included in the tables.

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What is known

Physicians who must confirm, exclude, and/or report suspected abusive head trauma have limited, validated, evidence-based tools that can inform their diagnostic reasoning.

What this study adds

The 7-variable clinical prediction rule described in this validation study incorporates the predictive contributions of completed abuse evaluations to facilitate patient-specific estimation of abuse probability in young, acutely head-injured children hospitalized for intensive care.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chiabu.2024.106799.

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